Social Network Analysis: A Survey on Process, Tools, and Application

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Due to the explosive rise of online social networks, social network analysis (SNA) has emerged as a significant academic field in recent years. Understanding and examining social relationships in networks through network analysis opens up numerous research avenues in sociology, literature, media, biology, computer science, sports, and more. Therefore, certain studies review and discuss some research verticals of SNA, such as viral marketing, information diffusion, clustering, link prediction, and so on, to provide background knowledge and understanding. These studies still lack the SNA process, tools, and practical aspects in multidisciplinary applications. Inspired by these facts, we have discussed the background, process, tools, and application of SNA. First, we have presented a detailed description of the SNA process. Thereafter, we presented a comparative analysis of SNA tools and languages. Finally, we have discussed the various applications corresponding to SNA research verticals.

CCS Concepts: • **Information systems** → *Social advertising*; **Social networking sites**; **Social networks;**

Additional Key Words and Phrases: Information diffusion, influence maximization, link prediction, community detection, social network analysis

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ACM 0360-0300/2024/04-ART192
https://doi.org/10.1145/3648470

ACM Reference Format:
Shashank Sheshar Singh, Samya Muhuri, Shivansh Mishra, Divya Srivastava, Harish Kumar Shakya, and Neeraj Kumar. 2024. Social Network Analysis: A Survey on Process, Tools, and Application. *ACM Comput. Surv.* 56, 8, Article 192 (April 2024), 39 pages. https://doi.org/10.1145/3648470

1 INTRODUCTION

A change in behavior in how individuals connect, communicate, and share their ideas, opinions, information, innovations, and so on, has been brought on by the growth of online social networking. This behavioral change in people leads to changes in relationships, business operations, marketing, elections, and more. Therefore, analyzing social networks has become necessary to understand and describe the social behavior of users and their relationships. To investigate the individual role, people's involvement, communities' roles, and so on, **social network analysis (SNA)** has replaced traditional social science methodologies including questionnaires, surveys, and interviews [100, 159]. The SNA can analyze a variety of practical aspects of multidisciplinary applications, including finding subject matter experts in a particular area, finding patterns in clickstreams on the web, finding faculty at different universities that have emerging communities of interest, mapping communities of expertise in various fields, mapping interactions among blogs on different topics, mapping an executive's network based on email flows, revealing how infections spread among patients and staff in the hospital, accelerating diffusion by identifying opinion leaders, and more.

The SNA has dealt with various research verticals such as influence maximization [153–155, 158], information diffusion [161], community detection [116], link prediction [1, 99, 156, 160], and more. Extensive research has been done considering these SNA research verticals. These research studies explore the impact of social relationships and structures to provide solutions for real-world, multidisciplinary applications. The SNA procedure is still difficult, though, because network data is growing so quickly. This data is available for researchers on an unprecedented scale and is unstructured, incomplete, and voluminous [145]. Therefore, to tackle these challenges, researchers need to keep developing social data science techniques. Although various developments have been made towards specific network analytics, capturing generalized knowledge from networked data collected from different sources remains an issue. Also, many analysis and visualization SNA tools and software were developed to analyze, understand, and explore network data [137]. The selection of a tool is still challenging, as each of these tools has different pros and cons.

Some of the surveys [100, 137, 145, 159, 161] present in the literature cover SNA understanding and knowledge. The authors of Reference [161] covered the influence maximization problem and information diffusion analysis. They have presented a detailed study of bench-marked methods and their comparative analysis. The authors of Reference [100] have presented a detailed survey on the link prediction problem and its applications. Some of the studies [137, 145] discussed the tools and software used for SNA analysis. The authors of Reference [159] presented SNA measures, network structure, language analysis in online networks, and applications. However, these studies are very specific to an SNA problem and have not discussed the SNA process and its multidisciplinary application corresponding to SNA research verticals.

SNA provides important insights into social interactions inside networks, opening up prospects for research in a number of fields, including sociology, literature, media, biology, computer science, and sports. While prior research has looked into particular SNA research fields including viral marketing, information diffusion, clustering, and link prediction, it frequently lacked a thorough knowledge of the SNA method, tools and real-world applications in multidisciplinary contexts. This article will examine the SNA attributes, methodology, tools, and applications of SNA in an

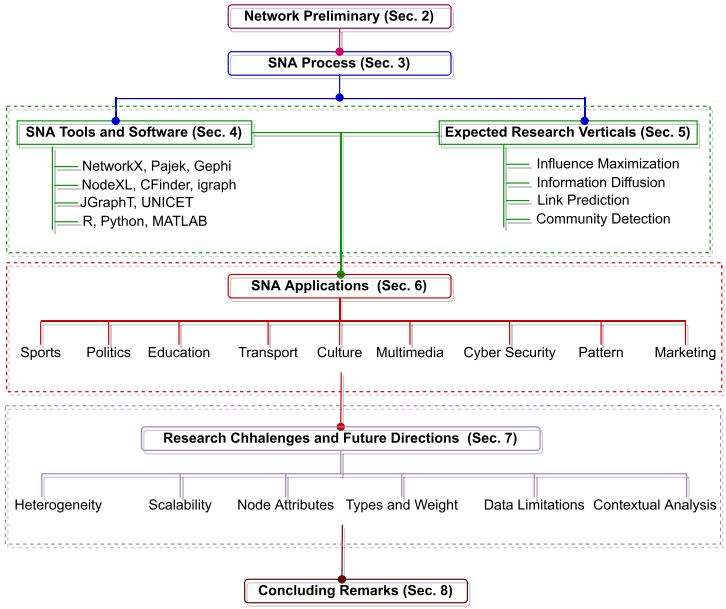


Fig. 1. Illustration of survey overview.

effort to close this gap. It gives a thorough explanation of the SNA procedure, contrasts different SNA tools and languages, and looks into applications connected to various SNA study fields. The following discussion covers the survey's primary contributions.

(1) This article first discusses the background information about social network modeling and network properties such as structural, relational, and individual. This preliminary knowledge will help to better understand network analysis.
(2) To comprehend the task involved with the analysis, the social network analysis process is presented. Then, we talked about the SNA software and tools for network investigation and visualization. Additionally, a comparison of these tools is provided.
(3) SNA research verticals such as information diffusion, clustering, viral marketing, link prediction, and so on, have also been briefly discussed.
(4) We have also covered some SNA multidisciplinary applications and their literature to provide a brief idea about the network analysis process and its uses.
(5) This survey also covers the difficulties and potential paths for SNA research.

The major goal of this survey is to serve as a jumping-off point for new social network analysis researchers. Regardless of the problem statement, the survey initially describes the history and characteristics of social networks before introducing the SNA procedure. The SNA process teaches users how to analyze social networks by posing research questions, gathering data, visualizing results, and so on. To have a better understanding, newcomers might use this survey as a starting point. The organization of the article is given in Figure 1. The preliminary knowledge, such as SNA modeling and network properties, is discussed in Section 2. Thereafter, we will provide a discussion on the SNA process and its phases in Section 3. Then, we discuss and compare SNA tools and software in Section 4. We present expected SNA research problems in Section 5. Next, we discuss multidisciplinary applications and their solution in Section 6. At last, we present a discussion on SNA research challenges and future direction in Section 7. Section 8 is devoted to the conclusion.

Table 1. Illustration of Network Properties

Network Properties	Network Measures	Social Facets		Measuring Factors	References
		Node	Edge		
Structural	Density	-	✓	Network Connectivity	[159]
	Size	✓	✓	Number of Node	[159]
	Diversity	✓	-	Node Type, Relationship Type	[159]
	Structural Holes	✓	-	Connectivity	[20]
	Cliques	✓	-	Completeness	[159]
	Degree	-	✓	Neighbors Connectivity	[155, 158]
	Betweenness	✓	-	Shortest Paths	[159]
	Closeness	✓	-	Shortest Distance	[159]
	Clustering Coefficient	✓	-	Triad	[1, 99]
	Isolates	✓	-	Independent Nodes	[65, 159]
	PageRank	✓	✓	Number and Quality of Links	[16]
	HITS	✓	✓	Hubs and Authorities	[133]
Relational	Tie Strength	-	✓	Time, Emotional Intensity, Intimacy, Reciprocal Services	[65]
	Accessibility	✓	-	Connectedness, Closeness	[161]
	Likeability	✓	-	Acceptability, Community Information	[65]
	Reputation	✓	-	Trustworthiness	[161]
	Trust	✓	-	Propensity to trust, Trust in others	[159, 171]
	Dependence	-	✓	Betweenness Centrality	[159]
	Expected Reciprocity			Personal/Professional Favors	[156]
	Competition	✓	-	Characteristic Matching, User's interest, Product Information	[100]
Individual	Personality	✓	-	Self-monitoring, Big-5, Feedback, Community	[5, 12, 164]
	Emotional Intelligence	✓	-	Closeness, Likeability	[112, 149]
	Intentionality	✓	-	Participation, Engagement	[196]
	Past Experience	✓	-	Historical Patterns (clusters, connectivity)	[178, 179]
	Sentimental Analysis	✓	-	Opinion, Intentions, Attitude	[152]

2 PRELIMINARY

2.1 Network Modelling

A social network is made up of the actors and links that connect them to form a social structure. The actors can represent persons, organizations, countries, companies, animals, words, web pages, families, topics, and so on, while their connections denote kinship, interactions, affiliations, affirmative, or other role-based relationships. A social network can be represented by a graph $G(V, E)$, where V and E denote nodes and edges, respectively, and where $|V| = N$ and $|E| = M$. The social network provides a new direction for analyzing these social actors for different applications. The SNA offers various methods to investigate and analyze multiple social structures, such as local structure, global connectivity, social impact, vital nodes, and so on, of social networks.

2.2 Network Properties

We will talk about the fundamentals of social networks in this section. Table 1 presents network properties with their measuring factors and social facets.

2.2.1 *Structural Properties.*

— **Density.** The network density calculates the proportion of nodes that are linked together or the number of connections that are actually made as opposed to all conceivable connections. The density $D(G)_D$ of a directed and un-directed network can be computed as $D(G)_D = \frac{2\times M}{N(N-1)}$ and $D(G)_{UD} = \frac{M}{N(N-1)}$, respectively.

— **Size.** The quantity of nodes a network contains determines the size of the network. Hence, the mathematical formulation for the size of network $S(G)$ is $S(G) = |V|$.
— **Diversity.** The diversity of a graph is defined by different subtypes of nodes found in the graph. For a simple graph, the diversity is calculated as the number of non-overlapping community structures found in a graph that shows different subgraphs of closely connected nodes within the original graph.
— **Structural Hole.** According to Burt's thesis [20], a person's location in a neighborhood or other social structure may provide them with benefits or disadvantages. According to the **structural hole (SH)** hypothesis, users who hold bridge positions between various communities benefit because they have influence over the main information dissemination channels. Structural holes have been used in different fields for various purposes, such as to estimate managerial performance, estimate more profitable collaborative relationships, and maximize information diffusion.
— **Clique.** A clique is a subset of undirected graph vertices in which every adjacent pair of vertices is different from the others. In other words, a fully induced subgraph of a graph G is a clique of that graph.
— **Degree.** The number of edges that are incident to a vertex determines its degree (or valency) in a graph.
— **Betweenness.** In a connected graph, there is always at least one shortest path between any two vertices that minimizes either the number of edges the path passes through (for unweighted graphs) or the sum of the edge weights. The betweenness centrality of a vertex is determined by the quantity of these shortest routes that pass through it. It shows the distance between neighboring nodes. For example, a node with higher betweenness centrality would be more influential on the network, since more data would pass through it.
— **Closeness.** Analyzing any social network is difficult due to the continual changes in the network. One of the criteria used to evaluate social networks is the degree of closeness between the nodes. The centrality of a network is measured mathematically by proximity. It is fairness's reciprocal. The reciprocal of the sum of the shortest paths between nodes is used to describe proximity mathematically.
— **Clustering Coefficient.** The clustering coefficient tells us how much a network's nodes tend to group together. The local clustering coefficient of a node quantifies how close together neighbors on a vertex must be to create a clique (full graph). The ratio of closed triplets to all possible triplets is the global clustering coefficient, however. Three nodes that are joined by two (open triplet) or three (closed triplet) undirected links are referred to as triplets.
— **Isolates.** Isolates are the nodes in the network that are not connected to any other nodes, i.e., the degree of such nodes is zero.
— **PageRank.** Based on the quantity and quality of inbound links a web page receives from other pages, an algorithm determines its PageRank score [16]. The basic idea is that a page with more high-quality inbound links is regarded as more credible and authoritative.
— **HITS.** Jon Kleinberg created the **Hyperlink Induced Topic Search (HITS)** Algorithm, a link analysis algorithm that assesses websites [133]. This method uses the web's link architectures to find and rank the pages that are pertinent to a given search. Hubs and authorities are used by HITS to specify a recursive relationship between web pages.

2.2.2 *Relational Properties.*

— **Tie Strength.** The quantification of the strength of any kind of relationship between two individuals is referred to as the tie strength of the bond between them. The current tie

strength technique was inspired by Granovetter's [65] weak connections theory and his four recommended aspects of tie strength: time spent engaging with someone, closeness level, emotional intensity level, and reciprocity level. Other aspects that contribute to tie strength between individuals are emotional support, social distance, and network structure.

— **Accessibility.** The flow of a network constantly meets resistance and sometimes becomes blocked for various reasons. Networks should be able to provide other routes when congestion causes the requirement for real-time rerouting. The ability of networks to provide an effective network flow from the source node to the destination node despite changes in network states is known as node-to-node accessibility.
— **Likeability.** In a restricted context, likeability refers to the number of nodes that are directly or indirectly connected to a specific network. The popularity or acceptability of a node in a network can also be predicted by likeability.
— **Trust.** When a person develops a good attitude toward a certain behavior, it is more probable that he or she will participate in that behavior. The belief forms the attitude, and the latter shapes the behavioral intention. The degree to which one side is ready to rely on something or someone in a certain scenario with a sense of relative security, even knowing adverse outcomes are likely, is known as trust [171].
— **Reputation.** Reputation is a complementary concept of trust. It is possible to think of reputation as a forecast of future behavior based on past contacts [171]. Correlating this concept with trust, a more trustworthy individual would have a higher reputation in his related community than a non-trustworthy one, which can also provide some guarantee of consistent future behavior.
— **Dependence.** Inferring causal activity relationships between network nodes is made possible by the dependency network technique, which also extracts causal topological relationships between network nodes. With the help of this idea, the network's overall dependence on each node may be estimated. Activities or node structure may be the sources of the dependence.
— **Expected Reciprocity.** Reciprocity is a quantitative metric used to examine complex networks. It quantifies how likely it is for two vertices in a directed network to be connected to one another. Reciprocity can uncover potential mechanisms and guiding ideas that could influence the topology of the observable network by spotting nontrivial patterns.
— **Competition.** In a situation of competition, competing networks contend for a limited number of nodes by altering the correlation between the nodes, the size of the network, and the connections between the nodes.

2.2.3 *Individual Properties.*

— **Personality.** Social Network Analysis over various networks is widely used to study the relationship among individuals in different groups, networks, communities (private and public), and so on. Jacopo et al. [164] investigate the relationships among various social networks platforms and personalities. The authors have focused on ego-networks, in predicting the Big-5 personality traits, which are agreeableness, conscientiousness, extroversion, neuroticism, and openness. Researchers have also worked in determining the specific personality traits based on job, position, nature of work, and so on. Araya et al. [12] propose a hypothesis of the Big Five personality traits of the presidents of Latin America. The hypothesis presents some important factors that act as a driving force for governing the states. Ahmed et al. [5] have worked to predict the Personality Traits and Mental Health of Bangladesh University students. They have focused on **Problematic Social Media Use**

(PSMU), which has caused various mental stress among professionals. The survey was conducted on 332 students, out of which 9% were found to be affected by PSMU.

— **Emotional Intelligence.** Social network analysis has provided wide platforms for both public and private networks. Users from communities, groups, social media accounts, and so on, can express their views on various platforms. A lot of public platforms, such as Quora, X (formerly known as Twitter), blogs, and so on, facilitate open discussions on various topics. Emotional Intelligence on various social network platforms has been studied by various researchers. Shardul et al. [149] discuss the understanding of Emotional Intelligence on social media by taking the insight from X. They analyzed 53,361 tweets from X and concluded that emotional intelligence tweets are mostly used by speakers, professionals, business organizations, and so on. Maria et al. [112] have done a survey on the negative consequences on adolescent mental health during COVID-19. They used a friendship network and conducted a small survey of 21 students in the 10th grade to analyze the negative impact of COVID-19 on mental health.

— **Intentionality.** With the increase in participation and influence of various communities over a digital platform, the investigation of the intentions of the users is high in demand. Intentional social actions are a wide area of research in the field of social network analysis. Celeb et al. [196] discuss the intentions of cyberbullies over social network analysis. They have proposed an AI Model for the automatic detection of cyberbullying over a large network. Their model is based on behavior and linguistic aspects used by the user over the network.

— **Past Experience.** This is sometimes also termed **HSNA (Historical Social Network Analysis)**. HSNA derives the conclusion based on the historical/past experiences of any social network analysis. Whetherell [179] discusses the concept of HSNA based on a broad range of activities based on place, time, and various results. Wellman et al. [178] do the SNA of historical communities to answer the present questions based on past experiences. Many questions such as the nature of family, community life, and connectivity among family/community were the groundwork done by them.

— **Sentimental Analysis.** It is the analysis of users' opinions, attitudes, feelings, intentions, and emotions of a user over a particular topic. Sentiment analysis is generally categorized into three broad categories: positive, negative, and neutral. Machine-based and lexical analysis are the most common ways to do sentiment analysis. Kumar et al. [152] present a review and comparative sentiment analysis of social media. Comparative surveys of different Machine learning models such as SVM, Naive Bayes, Linear Regression, and some feature extraction techniques such as POS, BoW, and so on, have been analyzed. The main challenge faced in sentiment analysis is the "Content vs. context" analysis. All AI models are generally based on content analysis but many times the content is affected by the context in which the content has been said/written. This leads to imporper interpretation of content in cases where the context is wrongly deduced. It is still one of the open challenges in sentiment analysis.

3 SNA PROCESS

Understanding and analyzing a set of users' relationships to identify important users and associations between users is the goal of social network analysis. To utilize SNA for various applications, the SNA method can determine the reach, activity, and influence of each person, group, relationship, and so on. As shown in Figure 2, network analysis focuses on a qualitative analysis, taking into account data source, data format, data volume, data analysis, and applicability. Data from online social networks are mostly divided into structured and unstructured categories. The majority of social network analysis relies on structured data (network topological data). Recent studies

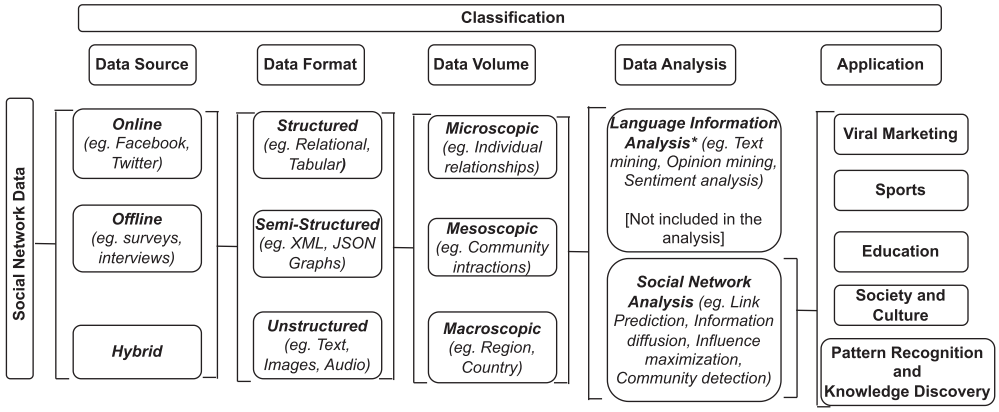


Fig. 2. Illustration of social network analysis (SNA) classification.

Table 2. Illustration of SNA Process

Step	Activities	Tools	References
Design	Identify boundaries, Clarify and design questions	UCINet, NetMiner, Cytoscape	[7, 66]
Data Collection	Surveys, Interviews, Online (Facebook, LinkedIn)	NVivo, Atlas.ti	[67]
Analysis & Visualization	Network construction, Descriptive analysis, Visualization enhancement	NodeXL, Ghephi, Pajek	[139]
Data Review	Validate; Look for questions	UCINet, Pajek, SPSS	[66, 139]
Prepare Evaluation	Match network results with context and stories	NVivo, Atlas.ti	[22]
Move into Actions	Weaving and other interventions	R, Gephi	[22, 138]

have, however, taken unstructured data into account. Unstructured contextual data serves as the foundation for language information analysis. As shown in Table 2, the SNA process for both types of analysis can be divided into the following six steps:

3.1 Design

The first task in the SNA process is to understand the problem statement and design research questions corresponding to the problem. It also identifies boundary conditions. For example, many research questions can be identified to predict missing links in online social interaction networks, such as information sources, network specifications and settings, and user behavior. How can features be identified or created based on available network information? Are there any dependencies between them? Similarly, an individual's impact and reach on the rest of the network can be considered a boundary condition for user influence.

3.2 Data Collection

Some software/tools/packages are available to complete network analysis tasks, such as data collection (spreadsheet software), data analysis (social network analysis tools), and data visualization (network visualization tools). The SNA outcomes are heavily dependent on the data's completeness, accuracy, and intelligence. Therefore, before data collection, researchers need to define the focus, which considers various elements such as focus cluster, individual, geographical location, catchment size, period, and so on. Then, we need to decide which relationships data (users, events, objects, organizations, etc.) we are going to use for analysis. The data can be collected from surveys, interviews, periodic audits, and online social networks (Facebook, X, LinkedIn).

Table 3. Comparative Analysis of SNA Tools, Software, and Programming Languages –I [137, 145]

Tools	Year	Open Source	Platform	Type	Uses	Language Used
NetworkX	2005	Yes	Linux, Windows	Library	Analysis & Exploration	Python
Pajek	1997	No	Windows	Stand-alone S/W	Analysis & Visualization	.NET
Gephi	2008	Yes	Mac, Linux, Windows	Stand-alone S/W	Graph Manipulation & Exploration	Java
Igraph	2006	Yes	Linux, Windows	Library	Analysis, Manipulation & Creation	C
JUNG	2010	Yes	Mac, Linux, Windows	Library & API	Analysis, Manipulation & Visualization	Java
AutoMap	2008	No	Windows	Stand-alone S/W	Text Mining	Java
SocNetV	2004	Yes	Mac, Linux, Windows	Stand-alone S/W	Analysis & Visualization	C++
UCINET	2001	No	Linux, Windows	Stand-alone S/W	Analysis & Visualization	BASIC/DOS
NodeXL	2008	Yes	Windows	Library	Network Discovery & Exploration	C#, .NET
CFinder	2005	No	Mac, Linux, Windows	Stand-alone S/W	Cluster Identification & Visualization	Java
JGraphT	2002	Yes	Cross-Platform	Library	Graph Visualization & Drawing	Java, C#
Hootsuite	2008	No	Linux, Windows, iOS, Android	Application	Social Media Management	PHP
R	1993	Yes	Mac, Linux, Windows	Language	Social Network Analysis	C, C++
Python	1991	Yes	Mac, Linux, Windows	Language	Analysis, Visualization & Text Processing	-
MATLAB	1984	No	Mac, Linux, Windows	Language	Analysis & Numerical Computing	C, C++, Java

3.3 Data Analysis and Visualization

The SNA analysis entails the research questions and boundary conditions developed at the start. There are many tools and software (such as NodeXL, Gephi, Pajek/UCINET, Igraph, Gephi, JUNG, AutoMap, CFinder, SocNetV, JGraphT, HootSuite, etc.), programming languages (such as Python, MATLAB, R, etc.) available for network visualization and analysis. So, there is no fixed way of analysis. However, some statistical analysis needs to be performed to answer some questions. For example, measuring centrality scores corresponding to each node can give an overview of the importance of individuals. The comparative analysis of SNA tools is listed in Table 3 and 4.

3.4 Data Review

Network analysis tells statistical information about the network data but does not explore any contextual details about findings. So, there is a possibility of misleading conclusions or incomplete details. Therefore, it is vital to analyze and validate the findings against contextual questions and operational experience. For example, data can be tested against some questions as follows:

— Do the analysis findings approximate with supervised data?
— Does any pattern seem unusual? Do issues in data support these unusual findings?
— Does the data satisfy known contextual information?

Therefore, reviewing data assures quality outcomes and opens new aspects to further explore the data.

3.5 Prepare Evaluation

Once data validation has been done, then match the data findings with the contextual information, stories, and assumptions we made during the design phase. Researchers also test performance metrics corresponding to research questions and context.

Table 4. Comparative Analysis of SNA Tools, Software, and Programming Languages –II [137, 145]

Tools	Network Layout					Network Measures						Multi User	Multi Relation
	Circular	Random	Star	F-R	K-K	Degree	Betweenness	Closeness	PageRank	HITS	Clustering		
NetworkX	-	-	-	-	-	-	-	✓	-	-	-	-	✓
Pajek	-	-	-	-	-	-	-	-	✓	✓	-	-	-
Gephi	-	-	-	-	-	-	-	-	-	-	-	-	✓
Igraph	-	-	-	-	-	-	-	-	-	-	-	-	✓
JUNG	-	-	-	-	-	-	-	-	-	-	-	-	-
AutoMap	-	✓	✓	✓	✓	-	-	-	-	-	-	✓	✓
SocNetV	-	-	✓	-	-	-	-	-	-	-	-	-	✓
UCINET	-	-	✓	-	-	-	-	-	✓	✓	-	-	-
NodeXL	-	-	✓	-	-	-	-	-	-	✓	✓	-	✓
CFinder	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	-	-	✓
JGraphT	✓	✓	✓	✓	✓	-	-	-	-	✓	✓	-	-
Hootsuite	-	-	-	-	-	-	-	-	-		-	-	-
R	-	-	-	-	-	-	-	-	-	-	-	-	-
Python	-	-	-	-	-	-	-	-	-	-	-	-	-
MATLAB	-	-	✓	✓	✓	-	-	-	✓	✓	✓	-	-

3.6 Move into Actions

Finally, weaving (process involves interpreting and making sense of the patterns, structures, and relationships) and other interventions can be performed.

3.7 Discussion

Despite extensive research on many areas of the SNA process, there is still a clear understanding gap regarding the function of data review and evaluation within this framework. The systematic examination and validation of data quality, dependability, and relevance at many stages of the SNA process, including design, data collection, analysis, visualization, and interpretation, has received relatively little attention. The majority of current research focuses on the technical components of SNA, such as community detection algorithms, centrality measurements, and network visualization tools. There are not many studies, nevertheless, that highlight the importance of data review and evaluation as a crucial step in the SNA workflow. Our capacity to guarantee the correctness, legitimacy, and generalizability of findings produced from SNA investigations is hindered by this research gap.

Finding the right data sources, choosing the right variables, and creating strict data-collecting protocols are frequently difficult tasks for researchers to complete during the design phase. Despite the importance of these choices, there is little information on how to develop reliable data collection frameworks and assess data appropriateness in the context of SNA. Furthermore, a wide variety of statistical and computational approaches are included in SNA's data analysis methodologies. The accuracy, validity, and potential biases of these methodologies have not yet been adequately evaluated in the study, though. Last but not least, adding assessment and review procedures to the SNA process can improve the accuracy, dependability, and reproducibility of research findings. Researchers' capacity to thoroughly evaluate the quality and reliability of SNA studies is hampered by the absence of standardized evaluation frameworks and norms.

Without a thorough analysis of the analytical techniques employed, researchers risk coming to the wrong conclusions or failing to notice crucial details in the network data. Additionally, social network visualization is essential for comprehending the intricate structures and patterns present in network data. The usefulness and suitability of these visualizations in accurately reflecting network features and enabling meaningful interpretations need to be evaluated, even though several visualization approaches have been established. Moreover, a thorough evaluation of the findings' robustness is necessary for the process of analyzing and assessing SNA data. It is yet understudied how network features, linkages, and emerging patterns can be verified using different techniques, such as qualitative data triangulation or comparison analysis. To progress the field of SNA and guarantee the integrity and validity of research findings, it is critical to fill the research gap mentioned above. Future research should concentrate on creating structured methods for reviewing and evaluating data at each stage of the SNA process, allowing researchers to use social network data to produce more thorough analyses and credible insights.

4 SNA TOOLS AND SOFTWARE

Web 2.0 has brought a revolution to the social network. Be it blogs, content sites, forums, wikis, search engines, and so on, all social sites have given a broad perspective towards sharing and managing the data over the social network. This has led to the concept of "Web Intelligence," which deals with the various dimensions of the web. With the wide use of the Internet around the world these days, a lot of people can be connected. Gartner [21] identified the significant development and growth of networks based on human behaviour, which further gives rise to a growing need for social network mining and social network analysis methods. SNA is required to have greater

insight into the network to study its evolution for applications such as community marketing, online shopping, recommendation mechanisms, and filtering. Since many upcoming technologies (wikis, digital bookmarks, online tagging, etc.) and services (GData, OpenSocial platforms, Facebook Beacon, Professional social networks, Google Friend Connect, etc.) have been proposed on the Internet, several effective SNA tools have been developed as depicted in Table 3 and 4.

There is a variety of tools available for analyzing and visualizing social networks, each with its own advantages and features. Social network data have been analyzed qualitatively and quantitatively using a variety of tools in the past 20 years. Researchers also use programming languages such as R, MATLAB, C/C++, JavaScript, and Python for SNA. Due to the fact that each tool has its own pros and cons, choosing the right tool for a specific task is still a challenge. Literature has published many papers on SNA tools. Four SNA tools—Pajek, Igraph, Gephi, and Network—have been compared by Camacho et al. [22]. A more detailed explanation of the above SNA tools has been given by Akhtar in 2014 in Reference [6]. Taking into account platform, graph layout, input file formats, type of network, graph characteristics, efficiency, and execution time of various features of graphs. Huisman and van Duijn [74] and APostolato [11] provided a comprehensive overview of various tools, but their comparative study was lacking. In contrast, Thangaraj and Amutha [169] have described the tools Gephi, NodeXL, and Pajek and elaborated on the factors that have led to the selection of these tools, including features, metrics, architecture, applications, input and output formats.

The article by Jain and Sinha [75] discusses a few social networking concepts, such as data-mining techniques and SNA tools. Shokeen and Yadav [151] described several social network centrality measures and five SNA tools (Pajek, NetDraw, NetworkX, Igraph, and Gephi). Lanny et al. [138] provide a list of several SNA tools, namely, GraphViz, Visone, SocNetV, NetworkX, Igraph, Gephi, JUNG, ORA, discussion graph tools, and R developed since 2000. They have pointed out that checking all SNA tools at once is one of the most difficult tasks. Akhtar and Ahamad [7] provided a comparative study of their five tools: R, NetMiner 4, Weka, KNIME, and RapidMiner. The research focused on comparing tools based on social mining concepts, their capabilities, platforms, and the input and output formats they support. Recently, Camacho et al. [22] conducted a scientometric study to discover application domains and active research areas in the field of SNA. They have also discussed SNA tools such as Neo4J, Pajek, Gephi, Cytoscape, Netminer, UCINET.

The most prominent approach in SNA is based on graph theory. Graph Mining tools are the most commonly used for analytically and visually investigating the social network. A number of tools are available to analyze the nodes and edges in a network. All the tools come with computing libraries, dependencies, web support (if any), application support (if any), and so on, and are referred to as SNA Software. GitHub [18], KDnuggets [83, 84], and Invovis [180] are a few websites that list around 70 SNA software. A few of them are listed below with names, reference material, brief descriptions, and license types.

— **SocNetV.** SocNetV. stands for Social Network Visualizer. It is used to analyze and visualize the social network. The features that make it a popular tool are that it is user-friendly, cross-platform, and freely available. It is licensed under GNU General Public License 3, formally known as GPL3. The software is supported in Windows, Linux, and MaC OS [80].
— **NetworkX.** Complex Network creation, manipulation, dynamics, and function can be done by using this tool. This software is made in Python language. The tool is mostly popular among data scientists, as it deals with different graph mathematics to build, analyze, and collaborate on different graph models [67]. It is licensed under BSD-new License.
— **Pajek.** Large networks are analyzed and visualized using this tool. The concept of a large network was first brought into account in November 1996. To analyze them, Pajek was

introduced and its very first version was launched in January 1997 and was presented at Sredin Seminar, Ljubljana, Slovenia. Since 1997, Pajek has been updated and a lot of versions have been launched as per the requirement of the users. The complete history with the version summary is listed in Reference [139]. It is Windows-based and is licensed under Free&Closed source.

— **Cytoscape.** This tool is designed to visualize molecular interaction networks. It is also used to integrate gene expressions and other state data. The tool is equipped with various plugins, such as new layouts, additional file format support, and so on. It was initially released in July 2002. It is written in Java and supports all the operating systems written in Java. It is licensed under GNU **Lesser General Public License (LGPL)**.
— **GraphX Apache Spark.** It is one of a component of Spark (Apache Spark is an open-source unified analytics engine for large-scale data processing which provides an interface for programming clusters with implicit data parallelism and fault tolerance) used for graph and graph parallel computation [183]. This tool introduces a new component, i.e., Graph abstraction. This tool is very much used in Facebook, LinkedIn connections, and so on. It is licensed under Apache's License.
— **JGraphT.** It is a Java library for graph theory. It deals with the various structures of the graph and their algorithms. This library supports different types of graphs such as directed, undirected, weighted, un-weighted, simple graphs, multigraphs, and so on. It is a powerful tool, as it supports various graph algorithms such as DFS, BFS, Clique detection, graph coloring, spanning, connectivity matching, and so on. It has dual license LGPL2.1 and **EPL2.0 (Eclipse Public License)** [115].
— **UCINET.** It is a tool that provides the facility of analyzing social networks as well as 1-mode and 2-mode data. This tool can read and write different text files and Excel files. It can handle a large number of nodes (32,767). The software can analyze various methods of centrality measures, role analysis, permutation-based statistical analysis, and elementary graph theory. Also, along with the aforementioned methods, the software has strong matrix analysis routines, such as matrix algebra and multivariate statistics, subgroup identification, and elementary graph theory [66].
— **Igraph.** This tool is used for creating and manipulating large networks. The tool is available in three languages: C, Python, and R, and also an interface for Mathematica. Researchers in the field of Network Science and related areas used it very much [46]. It was initially released in 2006, and the latest version came in September 2022. It is licensed under GPL 2.0.
— **JUNG.** It stands for **Java Universal Network/Graph** framework. It provides the facility of extendible language that can be utilized to model, analyze, and visualize different types of data in the form of graphs/networks [79]. This tool can also be used for annotating graphs, entities, and relations with metadata supported by various graph algorithms, such as; graph/network optimization, random graph generation, statistical analysis, and calculation of network distances, flows, and so on.

4.1 Discussion

Although SNA has received a lot of attention from a variety of sectors, there is still a need for research on how to evaluate and compare the numerous SNA tools that are frequently used, including Pajek, NetDraw, NetworkX, Igraph, and Gephi. There is not much research that systematically evaluates these tools' characteristics, capabilities, and usefulness for various study contexts, despite the fact that they provide functionality for network visualization, analysis, and modeling. Existing studies have mainly concentrated on individual tool descriptions, user guides, or particular applications; nevertheless, there is a dearth of thorough comparative research that discusses

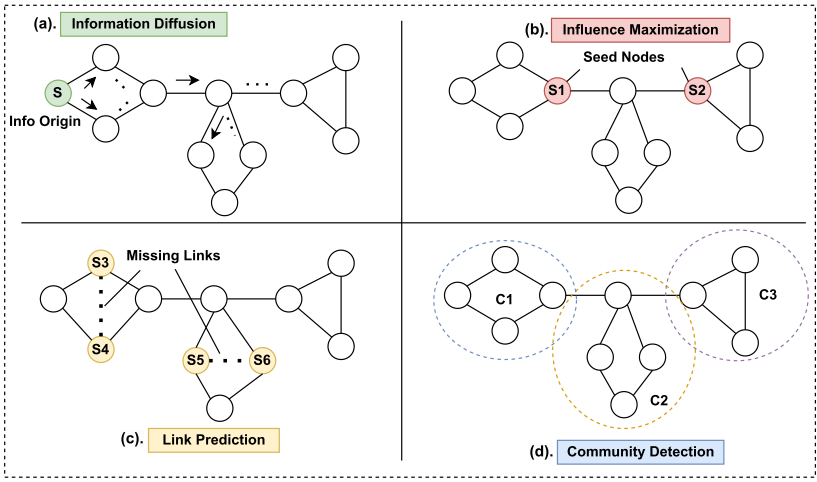


Fig. 3. Illustration of various SNA research verticals (a). Information diffusion: Source node S starts propagating information to its friends (neighbors), and friends share with their friends by word-of-mouth spreading, (b). Influence Maximization: nodes $S1$ and $S2$ are identified as seed nodes to maximize influence spread, (c). Link Prediction: Identifies missing links or future links such as $(S3, S4)$ and $(S5, S6)$, (d). Community Detection: Identifying communities such as $C1, C2$, and $C3$.

Table 5. Illustration of SNA Research Verticals

Research Verticals	Methods	Representations		
Information Dissemination	Threshold	Markov [174]	Partition [186]	Multi-Scale [109]
	Cascade	MINS [70]	DL [73]	SNICA [58]
	Epidemic	SIR [173]	SIS [25]	SIRS [107]
	Triggering	Greedy[85]	COEVOLVE [54]	ASYN [35]
	Time-aware	MIA [32]	CT-IC [90]	TCIM [8]
Influence Maximization	Approximation	Greedy [85]	CELF [104]	TIM [168]
	Heuristics	SIMPATH [64]	SPIN [126]	LDGA [33]
	Community based	CIM [140]	INCIM [17]	COFIM [148]
	Meta Heuristic	GA [19]	DPS [63]	Optimization [195]
Link Prediction	Similarity	SFCN [108]	CN [187]	Gc-LSTM [30]
	Probabilistic & Likelihood	HS [41]	SHOPI [99]	SBM [165]
	Dimensionality Reduction	PCA [24]	FFC [95]	CLP-ID [156]
Community Detection	Overlapping	Hypergraph-MT [43]	DPLPA [190]	ACNE-ST [29]
	Non-Overlapping	OTUCD [193]	GTAC [166]	NTCD [150]

the advantages, disadvantages, and distinctive characteristics of these SNA tools. For researchers, practitioners, and educators to choose the best tool for their unique study aims and data needs, it is imperative that they have a thorough understanding of the relative advantages and disadvantages of different tools. Additionally, there is a need for updated comparison studies that take into account recent upgrades, enhancements, and developments in functionality, because the majority of the research that has been done in this field has concentrated on earlier iterations of these tools.

5 EXPECTED RESEARCH VERTICALS

Research on social networks has received attention for the past two decades. The research on social media can broadly be divided into four parts, which are information dissemination, link prediction, influence maximization, and community detection, as shown in Figure 3 and Table 5. In the following subsections, we have tried to mention the state-of-the-art development in these domains.

5.1 Information Dissemination

Information diffusion (ID) is the word-of-mouth effect that happens online, and the message spreads from one place to another. Information diffusion can be categorized into different dynamics. The majority threshold model is one of the earliest and best-studied probabilistic diffusion models [172]. It was extended as the Independent Cascade Model [3] by Kempe et al. in 2003 [85]. The Shortest Path Model has been described by Kimura et al. to categorize the route of the dissemination [92]. The Unanimous Threshold Model is another important technique published in 2009 [31].

- **Threshold Model.** In this model, the collection of nodes determines whether or not they intend to forward the message. According to the linear threshold model [174], a node's choice to transmit data is determined by comparing its own threshold of influence to the combined influence of all nodes from which it has received packets. Recent studies have demonstrated a relationship between a node's adoption threshold and a parameter obeying a truncated normal distribution in the process of information transmission [109, 186].
- **Cascade Model.** According to the independent cascade model, each node has a certain independent probability of influencing its neighbors [70]. Recent papers use deep learning models and fuzzy control algorithms to enhance and optimize the information dissemination network model [58, 73].
- **Epidemic Model.** Contagious disease transmission, political campaigns, and the dissemination of information such as rumors and news were all significantly impacted by the epidemic process [101]. Based on the characteristics of the model, there are primarily three epidemic models: **susceptible infectious recovered (SIR)**, **susceptible infectious susceptible (SIS)**, and **susceptible infectious recovered susceptible (SIRS)** [87]. The SIR model assumes that every person in a community is susceptible, infectious, or disease-free and immune for the rest of their lives [173]. In SIS, susceptibles contract the disease by contact with infected people at a rate based on the number of infected contacts multiplied by the spread rate [25].
- **Triggering Model.** The generalized version of the threshold and cascading models is the triggering model, as shown in Reference [85]. Models for triggering enable effective simulation of interspersed diffusion and network events, and they produce traces that follow general diffusion and network patterns seen in real-world networks [54]. As a modification of the triggering algorithm, for each node to determine the final estimate, a diffusion sequential asynchronous state estimation algorithm has recently been presented in Reference [35].
- **Time-aware Model.** Time is not taken into account by the threshold, cascade, epidemic, and triggering diffusion models. When no more nodes are triggered, these models end the diffusion process. Some propagation models, however, must maximize the transfer of social influence over a given time period. They are referred to as time-aware models [32].

5.2 Link Prediction

Link prediction (LP) can be broadly defined as the task of predicting missing links in a network. In the case of static networks, which do not have a temporal component associated with them, it uses the current structure of the network to determine which links have a higher probability of

being created to ensure a complete design of the network. In the case of dynamic networks, link prediction is the task of predicting future links based on the previous pattern of changes in the network.

- **Similarity-based.** Similarity-based link prediction is computationally the simplest kind of link prediction category, in which similarity scores are calculated for node pairs that have to be evaluated for the possibility of link existence [30, 108, 187]. These methods can use information from either the immediate neighborhood of nodes themselves (local similarity-based) or they can also take the entire structure of the graph into account (global similarity-based). New methods that contain tradeoffs between these separate categories of information have also recently emerged (quasi-local similarity-based), and they show relative improvements in the overall task of link prediction.
- **Probabilistic & Maximum Likelihood-based.** For any given graph, probabilistic models evaluate a given objective function and optimize it based on different parameters of the graph [41, 165, 175]. Essentially, the probability of a new link is estimated using conditional probability, which gives an estimation of its presence on the overall graph-based objective function. A link that enhances the parameters of a graph such as overall connectivity is given more importance.
- **Dimensionality Reduction.** Link prediction techniques that are based on the techniques of network embedding and matrix decomposition come under the subset of dimensionality reduction [24, 95, 156]. This is because, at their core, these techniques transform the whole graph, which contains several nodes and edges into fixed length edges and node vectors that are further distilled to calculate link likelihoods. Embedding-based techniques extract node vectors from the original graph such that similar nodes that have approximately matching neighborhoods have small-dimension vector representations very close to each other. The embedding space for such tasks is much smaller than the original unrefined representation of graphs such as those in the adjacency matrix form.

5.3 Influence Maximization

Influence maximization (IM) is the problem of identifying a subset of influential individuals in the network. Influence maximization first came into existence for mining the network value of customers [50]. The problem was first formulated in Reference [86]. The influence maximization problem can be divided into four parts, which are an approximation, heuristics, community-based, and meta-heuristic.

- **Approximation-based.** The authors of Reference [85] were the first to present a greedy solution for identifying influential users for viral marketing. The greedy solution is best suited for the small-scale network, as it is not efficient due to Monte Carlo simulations. Therefore, The authors of Reference [104] provide an extension of the greedy algorithm by utilizing the submodularity property to reduce the complexity of these simulations. Tang et al. [168] have presented a near-optimal efficient algorithm for seed selection. These approximation algorithms identify better-quality seed users but are not scalable to large-scale social networks. These methods guarantee theoretical approximation with poor practical efficiency.
- **Heuristics.** To improve the efficiency of seed selection methods, many heuristics [33, 64, 126] have been proposed. These methods do not guarantee theoretical approximation but have more practical efficiency than greedy solutions. Authors of Reference [34] presented a degree discount heuristic based on node degree. Once a node is selected as a seed, the diffusion degree of each of its neighbors is reduced by one. This is because it is not able

to influence the seed node. The above mentioned degree discount heuristic method is an extension of highest degree seed selection method.

— **Community-based.** Some clustering-based methods [17, 140, 148] have also been presented in the literature to improve the efficiency of approximation algorithms by reducing the search space of seed nodes. Singh et al. [154] have presented a **context-aware community-based method (C2IM)** to identify seed nodes. They have utilized hierarchical clustering based on degree distribution to identify clusters. These clusters have been utilized to influence spread within clusters for seed selection. They also have incorporated the user's interest in the diffusion process. This method is highly efficient when the cluster label is known.

— **Meta-heuristics.** Recently, many meta-heuristic-based methods [19, 63, 195] have been also proposed for IM problem. These methods provide a tradeoff between influence spread and efficiency. The authors of Reference [155] presented a learning-based solution using a **particle swarm optimization (PSO)** technique to identify seed users. They have used learning automata to avoid local optimal solutions of the PSO strategy. Some other nature-inspired meta-heuristics also have been used for influence maximization, such as IM-SSO [153] and ACO-IM [157].

5.4 Community Detection

Community detection (CD) is a well-established technique in social network analysis. Communities, also known as clusters or modules, are collections of vertices that most likely have comparable traits and/or function in related ways inside the graph [57]. Community detection can be divided into mainly two subgroups, which are overlapping [49, 128, 181] and non-overlapping communities [26, 193]. When vertices belong to more than one community, then the boundaries of communities overlap and create overlapping communities. No node shares a common interest and exists in different communities in a non-overlapping community. A framework based on statistical inference is created in one of the more recent publications to describe the hierarchical structure of hypergraphs [43]. The technique enables the concurrent detection of overlapping communities in the presence of higher-order interactions, as well as the rational inference of missing hyperedges of any size. In Reference [190], the authors have created a label propagation method by exchanging information with fixed neighbors while maintaining the nodes' own labels. Another method establishes the community ownership of a node [150]. Some of the methods such as MINC-NRL [36], IEDC [68], GenPerm [26] can be used both for overlapping and non-overlapping community detection. Based on the strategy, it is possible to broadly categorize both overlapping and non-overlapping community detection techniques into the following groups.

In one of the publications [29], overlapping communities of vertices are modeled using an adversarial learning approach by mapping the relationship between communities and vertices into an embedding space. A non-overlapping community discovery task using graph convolutional networks is demonstrated in one of the manuscripts [193]. A framework based on game theory is presented for clustering using both the Louvain and Label Propagation methods [166]. By examining the connections between the node and its neighboring communities, community detection algorithms can be broadly divided into the following broad categories.

— **Graph Partition.** The graph clustering models divide a network into K different communities [57]. In their publication [88], the authors have provided a heuristic algorithm. A network is initially split into two arbitrary subgraphs, and nodes are optimized.

— **Spectral Clustering.** The network spectra in the spectral clustering method reflect the community structures [10]. The social network is divided using spectral clustering with perturbations on the regularized adjacency matrix and the normalized Laplacian matrix. The authors

of Reference [48] have combined communities in macroscopic and microscopic neural brain networks to produce clusters on the spectrum of normalized Laplacian matrices.

— **Dynamical Methods.** Communities are dynamically detected through random walks. The random walk computes node distances and community membership probabilities. The minimum length encoding method is used in information mapping [141]. The label propagation algorithms use a mechanism for information dissemination to identify diffusion communities [135].

— **Hierarchical Clustering.** This approach of hierarchical clustering uses dendrograms to identify hierarchical community structures. To find the clusters, they utilized divisive, agglomerative, and hybrid strategies. By gradually eliminating edges until a new community forms, the Girvan-Newman algorithm discovers community structure in a divisive manner [61]. An agglomerative method called fastNewman rapidly combines nodes, each of which is initially thought of as a community [127]. Using a hybrid approach, the community detection method based on structural similarity employs both agglomerative and divisive tactics [192].

— **Density-based Algorithms.** There are a few instances of this method, including **density-based spatial clustering with noise (DBSCAN)** [53], **structural clustering algorithm for networks (SCAN)** [184], and **locating structural centers for community identification (LCCD)** [176]. By assessing the density of entities, they can locate communities, hubs, and outliers.

5.5 SNA as a Big Data

In the age of big data, social network analysis has become a potent technique that allows us to understand nuanced patterns and insights within huge and convoluted networks of social interactions [52, 147]. The amount of data produced by online platforms, communication channels, and IoT devices has increased to previously unheard-of proportions in this era of digital connection [15, 144]. SNA makes use of this amount of data to unearth hidden connections, identify influential nodes, and highlight behavioral trends. SNA offers useful applications in areas such as marketing, epidemiology, and organizational management by applying advanced analytics to these enormous datasets. SNA also aids in our understanding of social systems. In essence, SNA as big data gives us the ability to navigate the complex web of human connections successfully and to fully utilize the potential of enormous data resources to obtain a deeper understanding of our globally interconnected environment [60, 98].

The authors of Reference [132] presented and evaluated an algorithm to support management decisions based on the network analysis processing of large-scale socioeconomic datasets gathered in accordance with the digital economy's tenets. The authors of Reference [170] examine the conceptual and methodological issues surrounding human activities in social networks, paying particular attention to the reliability and representativeness of large data analytics from social media. The overemphasis on X as a single platform, sampling biases resulting from hashtag selection, and nebulous and unrepresentative sampling frameworks are persistent issues. Big data social media interpretation is further complicated by the socio-cultural complexity of user behavior intended to achieve algorithmic invisibility. In another study [131], authors focus on the identification of latent patterns and trends, link events and information flows to pertinent and noteworthy events, and create a "clear" picture of the study topic and the important people it is connected to. The work helps to close the gap in the scientific literature created by the lack of development of practical issues involving the use of social network analysis to handle managerial tasks, despite the abundance of theoretical articles that outline the theory and methodology of such an analysis.

The authors in Reference [9] utilized centrality measures to find the closeness of two subscribers in the telecom network. The influencers were detected and targeted for viral marketing to increase

overall sales. In Reference [162], authors described the way state-of-the-art technological applications were permitting the experts to collect information with little bias. They also depicted novel challenges from the reality-mining techniques. One of the recent works [27] provided an overview of the development process, explored the consequences for management, and conducted an analysis of the subject matter. To elucidate six key aspects of management concerns, a demonstration was conducted showcasing an internally developed SocialNetwork API that encompasses six distinct functionalities. The findings can be extracted and subsequently presented through the use of data visualization techniques. Another study provided an analysis of the descriptive statistics pertaining to the many disciplines engaged in the publication of research on Big Data [72]. It examined the network indicators that reflect the interdisciplinary collaborations within these disciplines, including interdisciplinary communities and networks. It investigated the changes that occur within discipline communities over time. One of the recent studies showed the usefulness of including call networks as a novel Big Data source within the framework of positive credit information [197]. By employing a profit measure and profit-based feature selection, the inclusion of call networks has been shown to enhance profitability. A distinctive amalgamation of datasets, encompassing call-detail records, as well as credit and debit account information of clients, was employed to generate scorecards for individuals applying for credit cards. Table 6 presents a summary of the state-of-the-art SNA techniques concerning big data.

5.6 Discussion

Although there is a need for more research and improvement, areas including information diffusion, link prediction, influence maximization, and community detection have garnered a lot of attention. While previous studies in the field of information dispersion have examined diffusion models and mechanisms, there has been little investigation into the contextual variables that affect the dissemination of information in various social networks. The effectiveness of information dissemination tactics can be increased by investigating elements including network structure, content qualities, user behavior, and external events. These aspects have a significant impact on the dynamics of information dispersion.

Another crucial field of research is link prediction, which seeks to forecast future or absent links in a network. Although many different methods and algorithms have been put forth, there is still a research gap in thoroughly assessing and contrasting their performance. A crucial area of research continues to be the creation of reliable link prediction techniques that take into account a variety of network properties, including network architecture, node features, and temporal dynamics. Community identification, the process of locating closely knit communities or groups within a network, has also drawn a lot of interest. The development of reliable and scalable algorithms that can manage massive networks with millions or billions of nodes and edges, however, is still lacking in research. Additionally, there are still unanswered research problems about the incorporation of domain-specific constraints, the detection of overlapping communities, and the durability and quality of identified communities. Further investigation into the creation of novel models, algorithms, and evaluation procedures is required to fill up these research gaps. Table 7 presents the summary of the compared studies with respect to various criteria such as data format, volume, source, network type, research verticals, application domain, and so on.

6 SNA APPLICATIONS

Information diffusion, link prediction, influence maximization, community discovery, and other study focuses are only a few of the research verticals available in SNA. These verticals have a wide range of multidisciplinary applications, including the social sciences, education, literature, health, transport, and sports as well as the health and social sciences. SNA can be used to comprehend

Table 6. Comparision of the State-of-the-art SNA Techniques with Respect to Big Data

Study	Data Source	Application Domain	Analysis Scope	Problem Addressed	Big Data Technique	Limitations	Research Directions
[132]	X, Facebook	Socio-economic	Six-degree Separation	Monitoring system	SNA	Focus in data collection & processing	Incorporating spatial interactions
[170]	X	Socio-cultural	Hashtag analysis	Behavior analysis	Text mining	No group-norms in behavior analysis	Examining user behavior online & offline
[131]	X, Facebook	Geographic	Hashtags	Decision making	Clustering	Only node features are used	Incorporating more spatial dynamics
[9]	CRM, CDR	Telecom data	Centrality measures	Influential Nodes	SNA	Behavioral measures on multi-sim	Explored customers communities and fraud
[197]	Telco data	Financial	PageRank	Credit scoring	Ranking based	Correct profiling in data	Adding more credit bureau variables
[185]	S&P500	Stock Marketing	Degree	Community detection	Structural Analysis	Fairness	Adding robustness for outliers
[51]	LFR2000	Healthcare, Politics	Ensemble	Community detection	Structural Analysis	Fairness	Scalability
[191]	Mobile data	Commuting	Community indices	Community detection	Information extraction	Fairness and robustness	Adding fairness against network structure
[72]	Author network	Collaboration analysis	Network indicators	SNA analysis	SNA & Visualization	Only focus on Big data collaboration	Incorporating interdisciplinary collaboration
[27]	Facebook	Social Network	Management implications	Behavior analysis	Network visualization	Network specific API	Adding functions for X and LinkedIn

Table 7. Summary of the Compared Studies with Respect to Various Criteria

Study	Data Format	Data Volume	Data Source	Network Type	Research Verticals	Temporal Aspect	Application Domain	Challenges Addressed	Open Issues
[186]	structured	microscopic	online	directed	ID	static	email	influence of adoption thresholds	scale-free network
[70]	structured	microscopic	online	undirected	ID	static	academic coauthor network	Minimum-sized Influential Node Set selection	large scale network
[73]	structured	microscopic	online	directed	ID	static	media communication	fuzzy control algorithm	static network
[173]	structured	microscopic	online	directed	ID	static	Disease Modelling	disease control strategies	large size
[85]	structured	macroscopic	online	undirected	ID	static	co-authorship networks	natural greedy strategy	not dynamic
[54]	structured	macroscopic	online	directed	ID	dynamic	activity prediction	continuous-time model	node attributes are considered
[104]	structured	macroscopic	online	directed	IM	static	outbreak detection	cost-effective	not dynamic
[64]	structured	microscopic	online	directed	IM	static	collaboration network	linear threshold model, scalable algorithm	not dynamic
[19]	structured	microscopic	online	directed	IM	static	collaboration network	more diverse sets of feasible solutions	not experimented on different datasets
[108]	structured	microscopic	online	undirected	LP	static	discover future common neighbors	robust	not dynamic

(Continued)

Table 7. Continued

Study	Data Format	Data Volume	Data Source	Network Type	Research Verticals	Temporal Aspect	Application Domain	Challenges Addressed	Open Issues
[187]	structured	microscopic	online	undirected	LP	static	resource receiving process	path-independent method	not dynamic
[99]	structured	microscopic	online	directed, undirected	LP	static	finding missing link	similarity	not dynamic
[24]	structured	microscopic	online	undirected	LP	static	subgraph embeddings	e effectively eliminate noise	not dynamic
[43]	structured	microscopic	online	undirected	CD	static	structural organization of hypergraphs	robustness against addition of noisy hyperedges	not dynamic
[190]	structured	microscopic	online	undirected	CD	static	propagation-based overlapping community detection	preference selection strategy	networks without attribute informa-tion
[166]	structured	macroscopic	online	directed	CD	static	Non-Overlapping Community Detection	large network	not dynamic
[59]	structured	microscopic	online	undirected	centrality analysis	static	subject matter expert finding	Bibliometric Analysis using journal sources	not dynamic
[23]	structured	macroscopic	online	undirected	ranking algorithm	static	expert finding	email content	not robust
[188]	structured	microscopic	online	undirected	ML	static	expert finding	Wikipedia editing patterns	not taken complex scenarios

(Continued)

Table 7. Continued

Study	Data Format	Data Volume	Data Source	Network Type	Research Verticals	Temporal Aspect	Application Domain	Challenges Addressed	Open Issues
[123]	structured	microscopic	offline	undirected	CD	static	relationship among attributes	maneuvering pattern of digital resources	small datasets
[134]	structured	microscopic	offline	undirected	centrality analysis	static	performance analysis	student behavior analysis	small datasets
[189]	structured	microscopic	offline	directed	centrality analysis	static	student interaction	correlation and regression analysis	small datasets
[62]	structured	microscopic	offline	undirected	CD	static	performance analysis	Communication analysis of online communities	small and incomplete datasets
[163]	structured	microscopic	offline	undirected	structural equation modelling	static	performance analysis	impact of COVID-19 on higher education	small datasets
[121]	structured	microscopic	online	undirected	CD	dynamic	Collaborative Research	centrality analysis	only from India perspective
[119]	structured	microscopic	online	directed	CD	dynamic	game maker identification in Any Soccer Match	centrality analysis	small datasets
[113]	structured	microscopic	offline	undirected	network cohesion	static	passing networks in football	intra-team communication	not considered individual player's communication

(Continued)

Table 7. Continued

Study	Data Format	Data Volume	Data Source	Network Type	Research Verticals	Temporal Aspect	Application Domain	Challenges Addressed	Open Issues
[124]	structured	microscopic	online	directed	centrality analysis	static	quantifying the performance of a cricketer	player-vs-player information	not dynamic
[118]	structured	microscopic	offline	directed	CD	dynamic	Analysis of Fictional Characters	character extraction	small datasets
[102]	structured	microscopic	offline	directed	NLP	dynamic	Analysis of Fictional Character	survey on character extraction	small datasets
[40]	structured	microscopic	offline	directed	centrality analysis	dynamic	distinguish novels and adapted films	convergences and divergences between the two forms	small datasets
[47]	structured	microscpoic	online	undirected	NLP	static	keyword similarity identification	graph based keyword extraction	dissimilarity between different articles not considered
[44]	structured	microscopic	online	undirected	finding key users	static	knowledge discovery	Simplification Logic	not dynamic
[94]	structured	macroscopic	online	undirected	NLP	static	knowledge discovery	dataset creation	less refined
[14]	structured	macroscopic	online	undirected	NLP	static	knowledge discovery	large data	handling unreliable data
[122]	structured	microscopic	online	undirected	clustering	static	Fake news detection	detection of content sharing pattern	size of dataset

(Continued)

Table 7. Continued

Study	Data Format	Data Volume	Data Source	Network Type	Research Verticals	Temporal Aspect	Application Domain	Challenges Addressed	Open Issues
[111]	structured	microscopic	online	undirected	sentiment analysis	static	NLP	polarization of text	size of dataset
[120]	structured	microscopic	offline	undirected	routing	static	traffic	disaster management	complex scenario
[136]	structured	microscopic	online	undirected	CD	static	disaster management	extreme weather conditions	no use of machine learning
[143]	structured	microscopic	offline	undirected	routing	static	traffic	Traffic Routing	complex scenarios
[97]	structured	microscopic and macroscopic	online	undirected	NLP	static	crime	classification of crime data	overlapping

Here, ID: Information diffusion, IM: Influence maximization, LP: Link Prediction, CD: Community detection.

how diseases, behaviors, and information spread within social networks in the area of health. It allows researchers to pinpoint important people or organizations that are essential to information distribution and can guide public health intervention plans. SNA contributes to the understanding of group behavior, social movements, and organizational dynamics in the social sciences by offering insights into social structures, power relations, and social influence. SNA can help with the design of efficient educational interventions and can promote collaborative learning environments in the field of education by shedding light on information sharing, cooperation, and learning networks.

6.1 Education

- **Identification of Subject Matter Experts.** One of the frequent applications of social network analysis is the identification of the subject expert in any specialized area. In the manuscript cited in Reference [59], the authors and co-author network is mined utilizing the centrality metrics. Another publication has demonstrated how effectively subject experts may be discovered using a graph-based approach that accounts for communication patterns from various organizations [23]. To accurately identify editors who are qualified specialists in any specific area, the scientists have searched Google Scholar [188]. The dataset is used to build a machine learning classifier that is used to find new expert editors and evaluate the nature and scope of their work across Wikipedia by correlating these credentialed experts with their Wikipedia editing behaviors. In one of their articles, the authors describe how they used social network analysis and topical expertise to create a system for identifying significant individuals in support forums [125]. Based on sentiment analysis of their posts, the most influential users are then classified as having a positive or negative influence. Through the use of network analysis, one of the works has shown how the esports business is growing in its demand for skilled workers [81].
- **Analyzing Student Performance.** Online course has been portrayed by authors in one of their earlier works as a social network where students are connected via learning platforms [123]. The network has been mined based on a number of network-measuring characteristics to identify how the pupils are using these digital assets or services. The authors have examined instructional methods for utilizing social networks in both formal and casual learning in the research [28]. One of the manuscripts proffered research results related to social network-based technical education conception and implementation [134]. The work also demonstrated a novel method for evaluating students based on their behavior throughout the course and its validation in comparison to the conventional model of students' evaluation. One of the most recent studies analyzed how students interact in online discussions and looked into how that relates to how well students learned in an asynchronous course [189]. In another research, the authors compared the social network structure for measuring the individual and team performance of the student players in a multi-user team game [62]. In one study, the effect of SNAs as an e-learning platform on students' academic performance during and after COVID-19 is investigated [163].
- **Analysis Pertinence of Women Scientists and Institutions.** One of the articles examines the significance of top women-only colleges and universities in India, examining their effects and prospects in the field of research [121]. A social network is used to depict educational institutions of all stripes and the cooperative research projects they conduct. Researchers' productivity has been analyzed based on their gender by Cole and Zuckerman [42], who discovered consistent evidence of a gender gap in research production. Researchers have discovered a significant disparity between the publications of male and female academics throughout the course of their careers [182]. Additionally, gender bias is still present

in the distribution of research grants worldwide. Women scientists who are early in their careers apply for NIH funding less frequently due to gender bias [105].

— **Identifying Emergent Communities of Interest among Faculties.** Information may be exchanged and new ideas can spread based on the user interactions on the web. A significant area of interest in the study of SNA is a reflection on the development of those groups and their detection. The authors of a recent paper have attempted an in-depth analysis of the practice of grouping social network users into communities of interest and suggested a new classification of these methods [39]. The authors of a different study have demonstrated topic-based key member extraction using text mining and SNA approaches [106]. The scientists found that the patterning of interactions is a major predictor of a person being seen as a leader by other virtual collaborators based on the results of social network research [167].

6.2 Sports

— **Soccer.** In one of the manuscripts, the authors used a dynamic social network structure to depict the players and the ball traveling between them [119]. To differentiate the game designers from the matches as a case study, they examined a few random games from the FIFA Men's World Cup 2018. In one of the studies, SNA is used to analyze the passing networks that resulted in goals scored by winning teams throughout the knockout rounds of the 2016 European and COPA American football tournaments. The purpose of this study is to evaluate SNA as a technique for analyzing the composition of passes that result in goals and to look for differences in the passing networks of the nations of the two different continents that contributed to goals scored [114]. In a different study, players are asked to rate the amount of communication they received and the perceived effectiveness of that communication on performance using an intra-team communication tool. The framework is designed to comprehend the connection between members of a professional football team through passing and communication utilizing SNA [113].

— **Cricket.** An alternative ranking method based on social network indicators and their assessment in a composite distributed architecture has been suggested by the authors of one of the articles [142]. The ranking of players is discovered based on network measuring metrics after the authors examined the cricket community from the perspective of a social network. Through the use of machine learning algorithms, the rising cricket stars can be identified [4]. Another research compares common machine learning methods for predicting the winner of the T20 cricket world cup [13]. Using a unique accuracy metric, Random Forest emerged as the top machine learning algorithm among the produced models. The authors of one of the articles has provided a sophisticated way to evaluate the value of a batsman's runs or a bowler's wickets [124].

6.3 Society and Culture

— **Extracting Social Network for Literature.** A fresh notion to assess the traits of the protagonist and antagonist from the influential nodes based on the intricate graph has been put out in one of the manuscripts [118]. Additionally, the authors have introduced a community detection technique grounded in game theory that groups actors who have a lot of ties together. Recently published works have categorized literary works into several groups and offered a method for extracting fictional character networks [2]. Character networks, graphs that examine interactions and relationships between various characters in a literary work, are utilized in the works, which makes them especially pertinent to fiction.

— **Analyzing Adopted Films and Stories.** To investigate the differences in the character categorization within a story and its adapted film, one method has proposed an effective

method using various weighted graph centrality metrics, including degree centrality, betweenness centrality, closeness centrality, and edge contribution factor [40]. The manuscript has also looked at how the relationships between the characters vary depending on the time frames. The movies have been divided into many clusters according to their degree of resemblance by the scientists using social network analysis based on factors including genres, ratings, distributors, nationalities, sizes, and profitability. According to the results, ratings and nationality rather than genre are the main determinants of the motion picture market's clustering, and profitability differs significantly between groups [91].

— **Discovering News and Media.** To extract keywords from textual news stories, the authors have developed a graph-based method that makes use of term co-occurrence [47]. To assess the quality of newspaper articles published on any given topic, researchers have proposed a revolutionary graph theoretic method [117]. One of the papers focuses on creating a framework for establishing an influence network among users of a social network. [44]. It is important to keep in mind that influence networks typically have a group structure when studying them.

6.4 Patterns and Knowledge Discovery

One study offers a database of disruption patterns for the study of online social networks [94]. One of the studies offers an alternate technique called B-mine that employs SNA and a bitwise method to mine common patterns [78]. One study intends to use a network representation of bibliography data to extract knowledge using the capabilities of social network analysis tools [45]. Three key applications for opinion and pattern mining from social networks are presented in one of the more recent researches [14]. The authors have demonstrated a method for automatically identifying user mobility patterns, estimating the degree of political polarization of the general public, and finding engaging hashtags for social media discussions.

6.5 Politics

— **Fake News and Misinformation.** One manuscript examined user interactions in broadcasting networks using hyper-graphs and labeled news articles for the detection of false information [122]. The technique would stop the spread of misinformation on social media by stopping it from broadcasting. Recent research on fuzzy set-theoretic comparative analysis suggests that societies have varied views on how to distinguish between real news and fake news [129].

— **Election Campaign.** The Italian constitutional referendum that took place in 2016 is the subject of research that was presented in Reference [111]. The study examined the ways in which X users discussed the referendum in the weeks leading up to election day and expressed their intentions to vote. One recent study investigated and assessed the effects of reciprocity, intergroup interaction, and imitation on the spread of online political incivility [37]. Given YouTube's promotional algorithms, the authors have also demonstrated that political incivility online is a persistent and ongoing problem rather than a one-time incident. The films created or shared by the candidates of the major political parties during the general election campaign for the government of Spain in April 2019 are the subject of another study [110].

6.6 Tourism and Hospitality

A different study focuses on Muslim women as a customer category in an effort to offer a fresh perspective on the literature on social media and purchasing choices in the hospitality and tourism industries [89]. One report looks into the variables that influence users' trust in information about

travel, which is posted and obtained from social media or travel websites [93]. As the tourism industry is a field that is always expanding, another study aims to comprehend the kind of impact that Portugal's hotel industry can have on its customers [56].

6.7 Advertising and Marketing

Advertising and marketing are the most popular applications of SNA analysis, as most of the **influence maximization (IM)** algorithms have considered a digital marketing scenario to identify influential spreaders for advertising and marketing [85, 155, 158]. The importance of visual presentation in the processing of cause-related marketing messages on social networking sites has been demonstrated by a recent study [69]. The characteristics of social networking sites that have an impact on Iranian consumers' purchasing decisions have been studied by several researchers [55]. Utilizing current marketing and social psychology theories, the research constructs are measured.

6.8 Disaster Management

In any crisis scenario, a cooperative game theory-based solution is suggested to control traffic and reduce individual vehicles' waiting times [120]. Every vehicle plays the role of a player and works to raise the value of its reward. The payoff value depends on the vehicle's various characteristics, such as arrival time, speed, priority, and traffic density. Another paper suggests a hierarchical paradigm for decision-making in disaster management [103]. The paradigm would help in understanding how ideas spread and how people acquire opinions in social networks while taking community detection into account. Using cognitive behavior models, the suggested framework has been shown to capture the heterogeneity of people and assess its effects on the pace of information spread and the convergence of opinions.

6.9 Transport

Due to poor resource management, autonomous traffic control is a challenging task that will always exist. The theoretical underpinnings of a reservation-based scheduling algorithm for intelligent traffic management have been described in one work [143]. Due to the suggested approach's ability to avoid collisions, minimize delays, relieve congestion, prevent starvation, avoid deadlocks, and operate quickly, it can be used in any semi-autonomous environment. One recent study examined the research issues surrounding data processing for intelligent vehicular networks and was inspired to use graph processing technologies to solve vehicle issues [194]. In a different research, a context-aware trust estimation scheme is proposed that can enable roadside units in a vehicular edge network to offer real-time crowdsensing services in a reliable manner by leveraging information from reputable sources selectively [76].

6.10 Cyber Security

SNA methods can help analyze digital terrorism and cybercrimes and fight against them. In the age of the digital era, terrorists and criminals are effectively utilizing online social networks for fraud, crime, and terrorism activities by interacting with unknown people online. Therefore, SNA methods can help detect individuals, patterns, and activities related to crime and offense. The authors of Reference [97] presented an **NLP (natural language processing)**-enabled classification-based decision support system to analyze crime trends.

6.11 Multimedia

Most of the earlier SNA research focuses on text mining, i.e., the content shared in text format, and ignores other data formats such as image, video, and audio. This is possibly due to the complex nature of the analysis task, internet technological restrictions, and data sharing limitations.

However, internet connections and data rates have been improved nowadays, so multimedia data sharing is rapidly growing. Accordingly, the need and interest in analyzing multimedia data are also increasing. Some of the work [38, 77, 177] has been presented in the literature to analyze images, videos, and audio content.

7 RESEARCH CHALLENGES AND FUTURE DIRECTIONS

- **Heterogeneity.** One of the main difficulties in SNA analysis is data heterogeneity. The vast majority of different types of data, including photographs, video, music, blogs, text, clickstreams, emails, and more, make up social media data. This heterogeneous data is collected from multiple sources (known and unknown) and formats. Generally, network data collection is a loosely coupled bottom-up process [146]. However, some studies [71, 82, 96] presented integration techniques to consider multiple data sources under a single interface for data generation.
- **Scalability.** The scalability of online social networks is the other significant obstacle for SNA analysis. Generally speaking, scaling is a difficult and complex task, even if it has grown more critical as a result of the explosive growth of online social networks [130]. Therefore, ignorance of scalability can lead to inaccurate results.
- **Type and Weight of Relations.** The most widely used indicators of personal importance are centrality measures including degree, betweenness, and closeness. These metrics ignore the various connection strengths and instead concentrate on the user's position in the network and the number of connections. Sibling ties, for instance, are stronger than those in other social networks, despite the fact that they are not linked to sites such as Facebook and X. Researchers have long disregarded relationship types and weights. To address uncertainties brought on by direct connections, it is also crucial to consider weights and kinds.
- **Node Attributes.** Most studies consider network structure only and ignore node attributes for different research verticals. For example, gender and age group can be helpful in marketing and advertising, like the female customer will be more inclined to cosmetic products. SNA studies typically evaluate a user's importance based on their access to or reach within the rest of the network. However, social media users are actual people with unique traits and abilities. Therefore, these attributes are not replaceable with network structural properties.
- **Data Limitations.** Another major challenge in SNA analysis is data limitations. Data quality depends on these limitations, such as data completeness, missing data, unstructured data, data volume, rate of change, data availability, and so on. For instance, a user prefers to use a social media platform privately due to privacy concerns; therefore, the missing data point probability could be high compared to overall data.
- **Lack of Public Datasets.** Many online social network dataset sources are publicly available, consisting of different network structural information over different periods. These networks correspond to a specific time, product, geographical region, and so on. However, these networks ignore node attributes such as context, user interest, age, country, skills, profession, belief, and so on, which are vital to analyzing the user's importance corresponding to an application. Limitations of API usability for data integration from different social networks also increase researcher challenges.
- **Contextual Analysis.** Most of the earlier SNA research ignores contextual information. For instance, users' interests and product information were missing from the earlier viral marketing solution. However, researchers have recently started focusing on contextual features such as topical, time, location, semantics, and so on. Identifying varying contexts of ambiguous words from social media is still challenging.

— **Privacy Preserving.** Privacy is another major challenge for social media users as well as researchers. Having access to someone's data can be highly useful in the age of data. Complete social media privacy is practically unattainable, because it begins even before using social media. Therefore, maintaining privacy on social media is quite difficult for researchers.
— **Incorporating Topological Network Structure.** Network topologies such as star, mesh, circular, bus, and so on, are not popular in social network analysis. However, topological information can be beneficial for identifying key users, missing connections, similar objects, information flow, and so on. But identifying topological structures is very complex in large-scale networks when multiple topologies exist.
— **Opinion Evolution and Propagation.** Most social network studies consider users' positive relationships and ignore negative connections. All relations do not only have a positive influence on others but also negative. Therefore, ignoring negative influence in information dissemination can lead to incorrect results. Hence there is scope of incorporating not only both positive and negative opinions but also analyzing opinion changes over a period of time to improve existing information propagation methods.

8 CONCLUDING REMARKS

The process, resources, useful aspects, and importance of social network analysis in cross-disciplinary applications were examined in this manuscript. Understanding the SNA procedure and choosing appropriate tools and libraries for network relationship analysis will be made easier with the aid of this study. Primarily, a brief discussion about network modeling and measures is presented. An in-depth SNA process with a comparative analysis of SNA tools is studied. A comparative analysis of the literature corresponding to various research verticals is discussed. The study looked at how social network analysis may be used practically in various situations. The discussion has concluded with research problems and potential directions. Researchers who are new to social network analysis will gain better knowledge and a solid foundation from this study.

COMPLIANCE WITH ETHICAL STANDARDS

The authors declare no conflict of interest. No human or animal subjects have been studied in this article. This article surveyed social network analysis applications.

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Received 1 November 2022; revised 30 December 2023; accepted 31 January 2024